FEDERAL SPECIFICATION

CALCIUM HYDROXIDE; TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Calcium hydroxide covered by this specification is intended for use in the treatment of water, sewage and general use. (See 6.1.)

1.2 Classification.

1.2.1 Type and Grade.—Calcium hydroxide, as specified herein, shall be furnished in one type and in one grade.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

2.1 The following Federal Specifications, of the issues in effect, on date of invitation for bids, form a part of this specification:

P–S–536—Soap and Soap Products, General Specifications (Methods of Sampling and Testing).

RR–S–366—Sieves, Standard, Testing.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents, or from the GSA Regional Offices.)

2.2 Other publication.—The following publication, of the issues in effect on date of invitation for bids, forms a part of this specification:

Nongovernmental:

American Society for Testing Material, Book of Standards.

(The American Society for Testing Materials, Book of Standards, is published by the American Society for Testing Materials, 1916 Race St., Philadelphia 3, Pa.)

2.3 Specifications and publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 The calcium hydroxide shall be suitable for the purpose intended and as specified hereinafter (see 6.1).

3.2 The calcium hydroxide shall be in the form of fine white powder, free from lumps and hard caking in the container when received and shall comply with the following requirements:

3.2.1 *Calcium hydroxide content.*—Shall be not less than 90.0 percent by weight of $Ca(OH)_2$ when tested in accordance with 4.2.1.

3.2.2 *Particle size.*—Not less than 95.0 percent, by weight, shall pass through a No. 100 sieve and not less than 90.0 percent shall pass through a No. 200 sieve. (See 4.2.2.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—A representative composite sample shall be taken from each delivery,

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placed in a clean, dry container, sealed, and sent to the laboratory for test. The sample shall contain not less than 1 pound nor more than 10 pounds.

4.2 Tests.

4.2.1 *Calcium hydroxide content.*—The Standard Methods of Chemical Analysis of limestone, quicklime, and hydrated lime shall be used. (See ASTM Standards, method number C 25–47, rapid sugar method.) This method determines the percentage CaO. To convert this figure to percentage $Ca(OH)_2$, multiply by 1.321.

4.2.2 *Particle size.*—Sieve tests shall be made in accordance with the applicable portions of method 210.1 (Sieve Tests) Federal Specification P–S–536.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, commercial packaging will be acceptable.

5.2 Packing.—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name of the material, the quantity contained therein, the name of the manufacturer or contractor, specification number, stock number, and the number of the contract or purchase order. In the case of bags, the name of the contractor and the contract or order number, may appear on a suitable waterproof tag securely fastened thereto.

5.4 Caution label.—Individual containers of calcium hydroxide shall have the following caution label:

CALCIUM HYDROXIDE
WARNING! CAUSES SEVERE BURNS TO SKIN AND EYES

Avoid contact with skin, eyes, and clothing.

Do not take internally.

When handling, wear goggles or face shield. While making solutions, add Calcium Hydroxide slowly to surface of solution to avoid violent spattering.

In case of contact, immediately flush skin with plenty of water and wash with vinegar; for eyes, flush with plenty of water for at least 15 minutes and get medical attention.

6. NOTES

6.1 The material covered by this specification is commonly known as "Hydrated Lime". Where hydrated lime is required for structural purposes the user is referred to Federal Specification SS–L–351.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and should specify the packaging, packing, and marking.

6.3 Bid samples.—It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally, no samples will be necessary, prior to award, to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample, should be definitely stated, the specification to apply in all other respects.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the

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Government may have formulated, furnished, or in any way supplied the said drawings, specification or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 *Army, Navy, and Air Force.*

7.2.1 *Applicable specifications and other publications.*

Federal Specification:
UU–S–48—Sacks, Paper, Shipping.

Navy Department Specification:
General Specifications for Inspection of Material.[1]

Military Standard:
MIL–STD–129—Marking of Shipments.

(Copies of specifications and standard required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Inspection procedures.*

7.2.2.1 *(Air Force.)*—The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall furnish test reports in duplicate, showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

7.2.2.2 *Navy.*—For Naval purchases the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—Calcium hydroxide shall be packed in 50-pound lots in type 3 W paper shipping sacks conforming to Federal Specification UU–S–48.

7.2.3.2 *For overseas shipment.*—Calcium hydroxide shall be packed in 50-pound lots in type 2 X paper shipping sacks conforming to Federal Specification UU–S–48.

7.2.4 *Marking.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129, Marking of Shipments.

7.2.5 *Ordering data.*—Procurement documents should specify whether domestic or overseas packing and marking are required (see 7.2.3 and 7.2.4).

NAVY INTEREST: Md S Y

[1] For Navy purchases only.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price cents.

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